

Local Field Potentials in Parkinson's Disease: Recent Advances in Biomarker Identification and Adaptive Deep Brain Stimulation (2021–2026)

Executive Summary

Local field potentials (LFPs) represent the summed electrical activity of neuronal populations and have emerged as critical biomarkers for understanding Parkinson's disease (PD) pathophysiology and optimizing therapeutic interventions. This review synthesizes recent literature from 2021 to 2026, examining advances in LFP-based biomarker identification, particularly focusing on beta oscillations in the subthalamic nucleus (STN), their relationship to motor symptoms, and their application in adaptive deep brain stimulation (aDBS) systems. Key findings indicate that beta oscillations, particularly in the 13–30 Hz range, correlate with bradykinesia and rigidity, explaining approximately 17% of symptom variability. Emerging evidence highlights the importance of aperiodic signal components, multi-spectral state dynamics, and real-world chronic recording capabilities. The field is transitioning from static stimulation paradigms to closed-loop systems that dynamically adjust therapy based on neural biomarkers, promising more personalized and effective treatment for PD patients.

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1. Introduction

Local field potentials (LFPs) represent the summed electrical activity of neuronal populations and provide a critical window into synaptic and oscillatory dynamics across brain networks ¹⁴. In Parkinson's disease, a progressive neurodegenerative disorder characterized by motor symptoms including bradykinesia, rigidity, and tremor, LFPs have emerged as essential biomarkers for understanding disease mechanisms and optimizing therapeutic interventions. The advent of deep brain stimulation (DBS) technology, particularly with the development of sensing-enabled neurostimulators, has enabled chronic recording of LFPs from subcortical structures, most notably the subthalamic nucleus (STN), opening unprecedented opportunities for both scientific investigation and clinical application ¹, ¹⁵.

The past five years (2021–2026) have witnessed remarkable advances in LFP research in PD. These developments span from fundamental understanding of oscillatory dynamics to practical implementation of adaptive stimulation systems. LFP signals recorded from DBS electrode contacts have become primary candidates for extracting suitable physiomarkers for adaptive DBS (aDBS), which promises to deliver personalized, state-dependent therapy ². The ability to record LFPs in real-world settings outside clinical environments has further expanded the translational potential of these biomarkers ¹⁰, ²².

This review synthesizes recent literature examining LFP characteristics in PD, with particular emphasis on beta oscillations in the STN, their relationship to motor and non-motor symptoms, the effects of therapeutic interventions including DBS and dopaminergic medication, and the emerging field of adaptive neuromodulation. We focus on studies published between 2021 and 2026 that have advanced our understanding of LFP-based biomarkers and their clinical applications.

2. Beta Oscillations and Their Role in Parkinson's Disease

2.1 Characteristics of Beta Oscillations

Beta oscillations, typically defined in the 13–30 Hz frequency range, represent one of the most prominent and well-characterized features of STN LFPs in PD patients. These oscillations are often subdivided into low-beta (13–20 Hz) and high-beta (20–30 Hz) bands, which may have distinct functional significance ², ⁴. Local field potential signals show prominent oscillations in the beta frequency range in PD patients, and these signals are considered important biomarkers, especially for tremor and rigidity ⁶.

Exaggerated beta frequency oscillations in STN LFPs are correlated with bradykinesia and rigidity in PD patients, and suppression of beta oscillations following high-frequency DBS to STN correlates with augmentation of motor function ¹¹. Subthalamic beta power has been shown to strongly correlate with dopaminergic neurodegeneration in experimental models, providing additional evidence for the pathophysiological role of these oscillations ²⁵. The

spatial distribution of beta activity within the STN is not uniform; periodic beta power is maximal over more dorsal, putative sensorimotor territories, while aperiodic components show different spatial patterns 7 .

STN single units strongly synchronize to beta activity, particularly in areas of high beta power, demonstrating the cellular basis of these oscillations 27 . The magnitude of spike-triggered average LFP was greater within the STN compared to extra-STN structures, and significantly higher spike-triggered average magnitude was observed for areas of high beta peak power inside the STN 27 .

2.2 Correlation with Motor Symptoms

The relationship between beta oscillations and motor symptoms has been extensively investigated, with systematic reviews providing quantitative estimates of these associations. The most frequently reported associations are between UPDRS-III scores of hemibody rigidity and bradykinesia and measures of contralateral STN beta oscillations 2 .

Beta-based LFP signal features, including higher spectral power, larger spatial extent, longer bursts, and lower complexity, can explain approximately 17% of individual variability in bradykinesia and rigidity symptom severity 2 .

Subthalamic high-beta oscillation power positively correlated with bradykinesia-rigidity improvement following DBS ($r_s = 0.549$, $p < 0.0001$), and the distance to maximum high-beta power negatively correlated with improvement ($r_s = -0.452$, $p < 0.001$) 26 . High-beta power and its distance were significant predictors for bradykinesia-rigidity improvement, explaining 37.4% of variance 26 . Elevated STN low- and high-beta band power in PD patients has been linked to worse mobility quality of life, with those demonstrating worse mobility showing increased low-beta ($Z = -2.47$, $p = 0.012$) and high-beta ($Z = -2.26$, $p = 0.023$) band power 4 .

The relationship between beta oscillations and tremor is more complex. UPDRS tremor scores are typically not correlated with time-averaged spectral beta power but rather with transient suppression of beta power, and other frequency bands including theta, low-gamma, high-frequency oscillations, and tremor frequency itself 2 . This suggests that different motor symptoms may be encoded by distinct oscillatory patterns.

Beta frequency band measures showed variable correlation to clinical gait measures ($r = -0.50$ to 0.82) in a systematic review of gait-related LFPs 13 . Two studies found decreased beta power during gait, one found increased beta frequency peaks during gait, and one found higher beta power during dual-task gait than during single-task gait 13 . These findings indicate that beta oscillations are dynamically modulated by motor behavior and task demands.

2.3 Beta Burst Dynamics

Recent research has emphasized the importance of transient beta bursts rather than sustained oscillations. The duration of beta bursts is crucial, with long-duration bursts (>375 ms) potentially signaling bradykinesia and rigidity, while short-duration bursts are associated with good clinical scores 2 , 3 . In one study using adaptive time-domain segmentation, adaptively constructed segments were significantly longer in the OFF-medication state and shorter in the ON-medication state ($p < 0.001$) 3 . Clinical improvement was highly correlated with the difference in spectral entropy of beta bursts between OFF and ON states ($r = -0.83$, $p < 0.01$), and longer oscillations (duration ≥ 375 ms) were highly correlated with clinical improvement, suggesting a pathological role 3 .

The temporal dynamics of beta bursts are also informative. Levodopa reduced the occurrence rate and duration of low beta states, with greater reductions correlating with greater improvement in motor impairment 28 . Levodopa also favored the transition of low beta states to other spectral states 28 . These findings suggest that the balance between different spectral states, rather than beta power alone, may be critical for motor function.

3. Subthalamic Nucleus Local Field Potential Recordings

3.1 Recording Techniques and Methodologies

STN-LFP recordings can be obtained through various approaches, each with distinct advantages and limitations. Intraoperative recordings during DBS electrode implantation provide high-quality signals but are limited to acute settings 26 , 29 . Recordings from externalized leads in the immediate postoperative period allow for repeated measurements over days but are constrained by the microlesion effect and infection risk 3 , 9 . Chronic recordings from implanted pulse generators with sensing capabilities enable long-term monitoring in naturalistic settings but may be affected by stimulation artifacts and limited battery life 1 , 22 .

The influence of inter-electrode distance on STN LFP recordings has been investigated, with implications for adaptive DBS protocols 12 . Spatial differences in LFP characteristics have been documented, with periodic beta being maximal over more dorsal, putative sensorimotor territories, while the aperiodic exponent is relatively larger ventrally 7 . These spatial patterns inform biomarker selection and contact targeting for adaptive stimulation.

Signal processing approaches have evolved to capture the dynamic nature of LFP oscillations. Time-delay embedded hidden Markov modeling (TDE-HMM) has been used to identify transient spectral states with frequencies under 40 Hz, revealing multiple distinct states beyond simple beta oscillations 28 . Continuous wavelet transform, wavelet coherence, short-time Fourier transform, and power spectral density analysis have been employed to analyze spectral changes during motor tasks 18 . Decomposition of power spectral density into periodic and aperiodic components has revealed distinct temporal dynamics and functional significance of these elements 7 , 21 .

3.2 Spatial and Temporal Characteristics

The spatial organization of LFP activity within the STN reflects the functional topography of this nucleus. Periodic beta was maximal over more dorsal, putative sensorimotor territories, while the aperiodic exponent was relatively larger ventrally 7 . This spatial dissociation between periodic and aperiodic features may inform biomarker selection and contact targeting for adaptive stimulation 7 .

Temporal dynamics of STN LFPs are influenced by multiple factors including the microlesion effect, medication state, and circadian rhythms. Total LFP power and aperiodic parameters (offset and exponent) showed an inverted-U trajectory following electrode implantation, peaking at 31-90 days and declining by ≥ 12 months 7 . Periodic beta power increased from approximately 1-3 months onward and remained elevated at 6-12 months, leading to a higher periodic-to-total beta ratio in late windows 7 . These findings indicate that the microlesion initially suppresses oscillatory beta more than broadband activity, with a later relative prominence of the periodic component 7 .

Long-term consistency of LFP physiomarkers has been examined, with implications for chronic adaptive DBS systems 30 . Understanding the stability of these signals over months to years is critical for developing reliable closed-loop algorithms.

3.3 Real-World and Chronic Recording

A major advance in recent years has been the ability to record LFPs in real-world, naturalistic settings outside the clinic. Real-world LFP recordings in PD patients shed light on the neural mechanisms underlying motor symptoms, with recordings from the STN and globus pallidus internus illuminating the role of oscillatory synchronization in parkinsonian motor symptoms 1 . LFP peaks were readily detected in real-world clinical settings during routine clinic visits 4 .

The ability to record LFPs in PD patients with DBS therapy allows chronic tracking of disease-related neurophysiologic function, and disease or symptom-specific LFPs can be recorded outside clinical settings, enabling patients to capture LFPs corresponding to real-world events 10 . Patient-triggered out-of-clinic recordings have been implemented, allowing patients to mark specific events or symptoms and capture corresponding neural activity 10 .

Diurnal fluctuations of STN LFPs have been characterized over several days and nights in the home setting 22 , 23 . STN LFPs showed consistent fluctuation based on behavioral state, with LFP power higher during wakefulness than during sleep, and little overlap in magnitude between states 22 . Beta frequency LFPs showed a clear and consistent diurnal pattern, closely linked to time of day and activity state 23 . These fluctuations can be detected in the home setting using commercially available devices, even in patients treated with DBS for several years 23 .

4. Deep Brain Stimulation and Adaptive DBS

4.1 Effects of DBS on LFP Patterns

Deep brain stimulation profoundly alters LFP patterns in the STN and connected structures. Suppression of beta oscillations following high-frequency DBS to STN correlates with augmentation of motor function in PD patients 11 . STN-DBS reduces the power of mu and beta rhythms and enhances synchrony at the motor cortices, with the 'on' stimulation state correlating with a marked reduction in late theta (7.5 Hz), alpha (10.5 Hz) (mu wave), and beta (20 Hz) wave power, along with more synchronized and coherent brainwaves 24 .

The effects of DBS extend beyond simple power suppression. DBS modulation of cortical bursting activity amplified the brain signals' informational content regarding instantaneous movement changes 8 . M1 low beta burst duration correlated with movement cycle width, a relationship that dissipated with therapeutic DBS, while high gamma activity correlated with enhanced motor performance with DBS-ON 8 .

Computational modeling has provided insights into the mechanisms by which DBS restores network dynamics. DBS in the STN can restore dynamics of striatal networks, with modeling studies demonstrating that high-frequency

stimulation alters the rate function for synaptic currents and modulates population activity 11 .

4.2 Adaptive DBS Systems

Adaptive DBS (aDBS) represents a paradigm shift from continuous, open-loop stimulation to closed-loop systems that adjust stimulation parameters based on real-time neural feedback. LFP signals recorded from DBS electrode contacts are primary candidates for extracting suitable physiomarkers for aDBS 2 . LFP-based aDBS can effectively modulate clinical symptoms in PD patients 20 .

The development of aDBS systems requires identification of reliable control signals. Patient-specific, gait-related biomarkers have been identified in both STN and cortical areas at discrete frequency bands, serving as putative control signals for adaptive DBS to improve gait dysfunction 18 . Each patient had unique frequency bands where power values significantly differentiated gait events (heel strike and toe off for each foot), mainly found within the canonical theta and beta ranges 18 .

Challenges in aDBS implementation include the relatively small percentage of explained variability in outcome measures (approximately 17% for beta-based features), partly due to challenges in physiomarker detection and subjective symptom quantification 2 . More research is needed to compare algorithms for LFP utilization in DBS programming to find a simple, practical, and time-saving approach 5 .

4.3 Programming Optimization Using LFPs

Analyzing LFPs has the potential to assist or streamline DBS programming, with beta oscillations identified as a promising LFP biomarker 5 . However, knowledge gaps and challenges exist, especially in utilizing intraoperative LFPs 5 . Subthalamic LFPs have been used for selecting chronic DBS contacts, with novel implantable neurostimulators allowing real-time LFP recording 19 .

Intraoperative subthalamic nucleus LFP recordings have been utilized to guide postoperative DBS programming 29 . The guiding effect of LFP during DBS surgery for programming has been investigated, with studies identifying LFP biomarkers correlated with improved motor symptoms after DBS 34 . Intraoperative LFP beta power and three-dimensional neuroimaging mapping predict long-term clinical response to DBS 31 .

Strong beta peaks captured at initial programming and their diminution correlated with symptom improvement, suggesting that identification, tracking, and modulation of LFPs correlate with subjective and objective clinical improvement 15 . Current and future applications of LFP-guided programming with sensing-enabled neurostimulators continue to expand 32 .

5. Biomarker Identification and Clinical Applications

5.1 Motor Symptom Biomarkers

Beta oscillations remain the most established biomarker for motor symptoms, particularly bradykinesia and rigidity. However, recent research has revealed a more nuanced picture involving multiple frequency bands and signal features. Beyond beta power, the duration and complexity of beta bursts provide additional predictive value 2 , 3 . A multivariate model considering all LFP states and their features predicted 50% of the variance in patients' hemibody

impairment OFF medication and in the change following levodopa, significantly outperforming models using only beta bursts (16% variance) 28 .

For gait dysfunction, LFPs are associated with and modulated by gait in PD patients, with potential to be used as a biomarker of gait dysfunction 13 . Beta frequency band measures showed low-to-strong correlation to clinical gait measures, though the relationship is complex and task-dependent 13 .

Aperiodic 'noise' elements of LFP, specifically the slope of the 1/f broadband, predict PD motor symptoms and outcomes of STN-DBS 17 . The aperiodic exponent of STN-LFPs reflects excitation/inhibition balance and might be a candidate biomarker for adaptive DBS 21 . Higher aperiodic exponents in STN LFPs reflect defined changes in basal ganglia network activity, tallying with lower levels of STN neuron firing and a balance tipped towards inhibition 21 . In PD patients, higher exponents accompany dopaminergic medication and DBS of STN 21 .

5.2 Non-Motor Symptom Biomarkers

While motor symptoms have dominated LFP research, emerging evidence indicates that non-motor symptoms are also reflected in LFP recordings. Non-motor symptoms such as impulse control and depression have been reflected in LFP recordings 2 . Low-frequency activity in the STN, particularly in the 4-12 Hz range, informs about the acute neuropsychiatric state in PD 16 .

Acute low neuropsychiatric state (OFF-medication) was associated with elevated theta/low-alpha power, with 6-8 Hz activity indicative of neuropsychiatric state change following medication intake 16 . These findings were most evident in ventral contacts closer to the limbic STN 16 . STN low-frequency activity may serve as a biomarker for acute neuropsychiatric state and neuropsychiatric responsiveness to dopamine 16 .

Basal ganglia LFPs have been investigated as potential biomarkers for sleep disturbance in PD 33 . STN LFP fluctuations are useful measures to distinguish between sleep and wakefulness in PD, with consistent fluctuation based on behavioral state 22 . Beta power was consistently higher during wakefulness than during sleep across all subjects, with little overlap in magnitude between states 23 . A statistically significant positive correlation was found between the beta frequency peak (Hz) and the scaled LFP difference between sleep and wakefulness ($r = 0.602$, $p = 0.008$) 23 .

5.3 Aperiodic and Multi-Spectral Features

A major conceptual advance in recent years has been the recognition that LFP signals contain both periodic (oscillatory) and aperiodic (broadband) components, each with distinct functional significance. Decomposition of power spectral density into periodic and aperiodic components has revealed that these elements have different temporal dynamics and spatial distributions 7 , 21 .

The aperiodic exponent of subthalamic field potentials reflects excitation/inhibition balance in Parkinsonism 21 . Higher aperiodic exponents tally with lower levels of STN neuron firing and a balance tipped towards inhibition 21 . This aperiodic component predicts motor severity and stimulation response 17 , suggesting it may be as important as traditional oscillatory biomarkers.

Multiple spectral states beyond beta have been identified in STN LFPs, including theta, alpha, and high beta bands, which behave oppositely to low beta states, increasing in occurrence rate and duration with levodopa 28 . These additional LFP states distinguished in the theta, alpha, and high beta bands provide complementary information about motor state 28 . The balance between competing spectral states in the STN is linked to motor impairment in PD 28 .

6. Future Directions and Challenges

The field of LFP research in PD is rapidly evolving, with several promising directions and persistent challenges. A key priority is improving the predictive power of LFP-based biomarkers. Current beta-based features explain only approximately 17% of symptom variability 2 , leaving substantial room for improvement. Multi-modal approaches combining multiple frequency bands, aperiodic features, and potentially other modalities (such as cortical recordings or kinematic data) may enhance predictive accuracy.

The development of robust, clinically practical aDBS algorithms remains a central goal. While proof-of-concept studies have demonstrated feasibility, more research is needed to compare different algorithms and identify optimal control strategies 5 . Challenges include accounting for inter-individual variability, adapting to changing disease states over time, and managing the complexity of multi-symptom optimization.

Real-world validation of LFP biomarkers is increasingly important. While laboratory studies provide controlled conditions for investigating LFP-symptom relationships, the ultimate test is performance in naturalistic settings with all their variability and complexity 1 , 10 , 22 . Patient-triggered recordings and chronic home monitoring will be essential for this validation.

The relationship between LFPs and non-motor symptoms requires further investigation. While initial studies have identified associations with neuropsychiatric state 16 and sleep-wake cycles 22 , 23 , the full spectrum of non-motor symptoms and their neural correlates remains to be characterized. This is particularly important given the substantial impact of non-motor symptoms on quality of life.

Technical challenges persist, including managing stimulation artifacts in chronic recordings, optimizing battery life in sensing-enabled devices, and developing signal processing approaches that are robust to non-stationarities and individual differences. The influence of factors such as electrode location, disease duration, medication state, and stimulation parameters on LFP characteristics must be better understood and accounted for in biomarker development.

Finally, there is a need for larger, multi-center studies with standardized protocols to validate findings across diverse patient populations and clinical settings. Many current studies involve relatively small sample sizes and single centers, limiting generalizability. Collaborative efforts and data sharing will be essential for advancing the field.

7. Conclusion

Local field potentials have emerged as powerful tools for understanding Parkinson's disease pathophysiology and optimizing therapeutic interventions. Research from 2021 to 2026 has substantially advanced our knowledge of LFP characteristics, particularly regarding beta oscillations in the subthalamic nucleus, their relationship to motor and non-motor symptoms, and their application in adaptive deep brain stimulation systems.

References (34)

Key findings from this period include: (1) beta oscillations, particularly in the 13-30 Hz range, correlate with bradykinesia and rigidity, explaining approximately 17% of symptom variability, with burst duration and dynamics providing additional predictive value; (2) real-world local field potential recordings in patients with deep brain stimulation for Parkinson's disease severity and stimulation response; (3) multiple spectral states beyond beta contribute to motor function, with levodopa-induced shifts in the balance between these states predicting clinical change with high fidelity; (4) real-world chronic recording is feasible and reveals consistent diurnal patterns linked to behavioral state; and (5) LFP-guided programming and adaptive DBS systems show promise for personalizing therapy.

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Despite these advances, challenges remain. The relatively modest predictive power of current biomarkers, inter-individual variability, and the complexity of multi-symptom optimization require continued research. The field is moving toward multi-modal, personalized approaches that integrate multiple signal features and adapt to individual patient characteristics and changing disease states.

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The translational potential of LFP research is substantial. As sensing-enabled neurostimulators become more widely available and algorithms mature, LFP-based adaptive DBS may become standard clinical practice, offering more effective and personalized treatment for Parkinson’s disease. The insights gained from LFP research also deepen our fundamental understanding of basal ganglia function and dysfunction, with implications extending beyond PD to other movement disorders and neurological conditions. The next five years promise continued rapid progress in this dynamic and clinically important field.

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